

High Performance PID Control of a Cascade Tanks System as an Example for Control Teaching

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Abstract—A detailed analysis of PID level control in the second tank of the cascade of two tanks is performed with respect to both, load disturbance attenuation and set-point change. Approximate formulas for extrema of time responses and for certain performance indices are derived, giving guidelines for controller settings. A simple method of choosing controller parameters is proposed that bases on time scale separation. For a reasonable transfer from one operating point to another, under control signal limitations, command signal generators with a feed-forward from the reference are proposed and the effect of anti-windup controller augmentation is examined.

I. INTRODUCTION

A model of a water tank can be found in almost every textbook, [6], [7], [8], [12] on control engineering. Classical P, PI and PID control for systems of water tanks like the one depicted in Fig.1 is the subject of many control teaching laboratory experiments, both physical[2], [1] and virtual [11]. As a result, every year hundreds if not thousands of students all

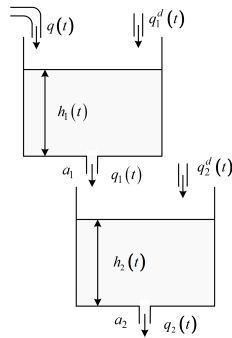


Fig. 1. Schematic diagram of a two tanks cascade system

over the world get in touch with this problem. Unfortunately, despite its popularity and importance for control teaching, its theory seems to be poor. The most advanced approach to be found in the literature [2], [15], based on pole placement which neglects the system zeros, does not allow to determine important control system properties. As a result, no appropriate tuning rules are available, and the controller tuning is to be completed experimentally. Unfortunately, there is neither appropriate guidance on how to do it, nor it is known what is the performance limit. In consequence even poorly tuned controllers may appear to be tuned properly.¹ The aim of the paper is to fill this gap with respect to a two tanks

cascade system. We also propose certain solutions to cope with large set-point changes that include not only anti-windup, but also model following, which can be accomplished using time-variable reference from a reference signal generator and possibly a feed-forward signal based on reference signal.

II. MODEL DESCRIPTION

In this section equations of tanks systems known from the literature will be written using normalized variables

$$u(t) = \frac{q(t)}{q_N}, x_1(t) = \frac{h_1(t)}{h_{1N}}, x_2(t) = \frac{h_2(t)}{h_{2N}}, \quad (1)$$

where q_N , h_{1N} and h_{2N} are certain nominal values that fulfill appropriate conditions of equilibrium.

A. Basic equations

The double tank system depicted in Fig.1 is described by the following set of equations

$$A \frac{dh_1(t)}{dt} = q(t) - \kappa_1 \sqrt{h_1(t)} + q_1^d(t) \quad (2)$$

$$A \frac{dh_2(t)}{dt} = \kappa_1 \sqrt{h_1(t)} - \kappa_2 \sqrt{h_2(t)} + q_2^d(t) \quad (3)$$

with

$$\kappa_1 = c_d a_1 \sqrt{2g}, \kappa_2 = c_d a_2 \sqrt{2g}, \quad (4)$$

where A denotes the cross-sectional area of tanks, a_1, a_2 denote cross-sectional areas of outlet orifices, c_d the coefficient of discharge, $h_1(t)$ and $h_2(t)$ liquid levels and $q(t)$ the input flow. Let us denote h_{1N} and h_{2N} nominal values of liquid levels in tank 1 and 2 respectively that fulfill

$$q_N = \kappa_1 \sqrt{h_{1N}} = \kappa_2 \sqrt{h_{2N}}, \quad (5)$$

from which it follows

$$h_{1N} = \frac{1}{\lambda} h_{2N}, \text{ where } \lambda = \left(\frac{a_1}{a_2} \right)^2. \quad (6)$$

Then the normalized equations are

$$\frac{1}{\lambda} T_N \frac{dx_1(t)}{dt} = u(t) - \sqrt{x_1(t)} + d_1(t) \quad (7)$$

$$T_N \frac{dx_2(t)}{dt} = \sqrt{x_1(t)} - \sqrt{x_2(t)} + d_2(t) \quad (8)$$

where

$$T_N = \frac{A h_{2N}}{q_N} = \frac{A}{c_d a_2} \sqrt{\frac{h_{2N}}{2g}} \quad (9)$$

¹A detailed analysis of PI control for a single tank system is given in [5].

and $d_i(t) = q_i^d(t)/q_N, i = 1, 2$. Figure 2 shows water levels when filling up the tanks and then draining them. It is interesting that when the input flow equals to zero then the upper tank becomes empty earlier than the lower one.

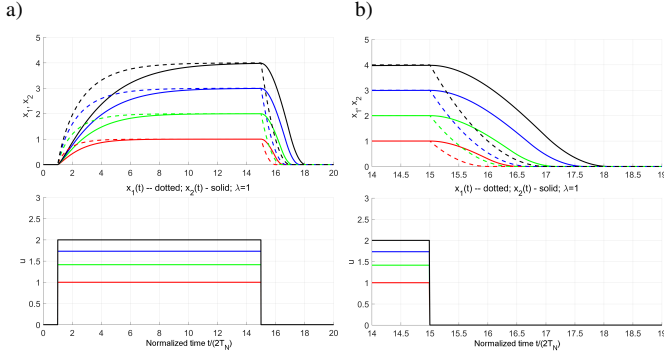


Fig. 2. a) Water tank filling up and draining to zero levels, b) zoom

B. Linearized model

Assuming $u(t) = u_0 + \Delta u(t)$, $x_1(t) = x_{10} + \Delta x_1(t)$ and $x_2(t) = x_{20} + \Delta x_2(t)$ the normalized equations are then linearized around a working point determined by u_0, x_{10} and x_{20} which fulfill the equilibrium equation

$$u_0 = \sqrt{x_{10}} = \sqrt{x_{20}}. \quad (10)$$

Linearization of these equations leads to the transfer function

$$\frac{\Delta Y(s)}{\Delta U(s)} = \frac{2\lambda u_0}{4T_N^2 u_0^2 s^2 + 2T_N(1 + \lambda)u_0 s + \lambda} \quad (11)$$

$$= \frac{2u_0}{(T_1 s + 1)(T_2 s + 1)}, \quad (12)$$

where

$$T_1 = \frac{2T_N u_0}{\lambda}, \quad T_2 = 2T_N u_0 \quad (13)$$

and

$$\gamma = \lambda \text{ for } a_1 \leq a_2, \quad \gamma = \frac{1}{\lambda} \text{ for } a_1 > a_2. \quad (14)$$

The value of this coefficient varies in extreme cases from 0 for the first-order system to 1 for the the second-order one with identical time constants.

C. Model parametrization and properties

From the control point of view, systems to be controlled are characterized by the ratio $\gamma = \min\{T_1, T_2\} / \max\{T_1, T_2\}$ where $0 < \gamma \leq 1$ for double tank system. This allows for a temporal characterization in which time scale is related² to $2T_N$, and for $u_0 = 1$ the system will be described by the following transfer function

$$K_p(s) = \frac{2d}{(s+1)(s+d)}, \quad d = \frac{1}{\gamma}. \quad (15)$$

²This can be attained using another time variable $t' = t/2T_N$ and related Laplace transform variable $s' = 2T_N s$, but we retain t and s in order not to complicate the notation

For $u_0 \neq 1$ there is

$$K_p(s) = \frac{2du_0}{(u_0 s + 1)(u_0 s + d)}, \quad d = \frac{1}{\gamma} \quad (16)$$

Exemplary characteristics of the upper tank, and of the lower one described in (16), are displayed in Figures 3 and 4. It is interesting to notice that while the high frequency gain of a single tank does not depend on the working point defined by u_0 , the high frequency gain of the cascade tanks system does depend on that point.

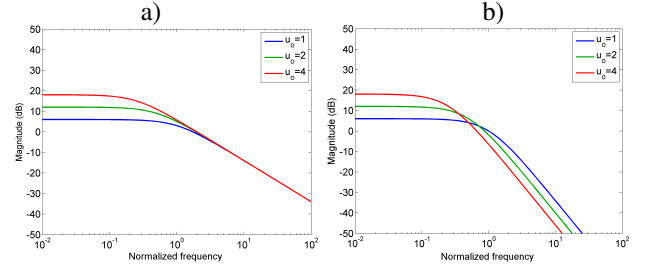


Fig. 3. Bode plots for the model linearized at different values of u_0 ; a) first tank, b) cascade tanks

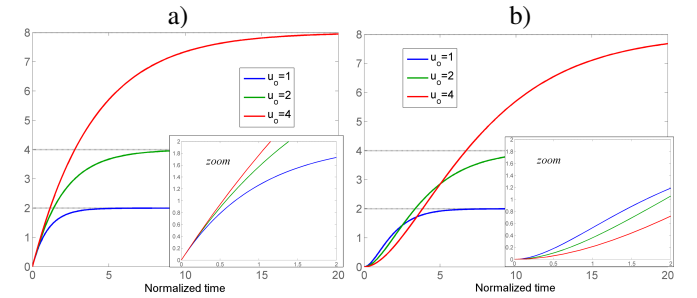


Fig. 4. Step responses of linearized model for different values of u_0 ; a) single tank, b) cascade tanks

III. CONTROLLER DESIGN

Let us consider a control system depicted in Fig.5 with a PID controller. The ideal controller with the transfer function

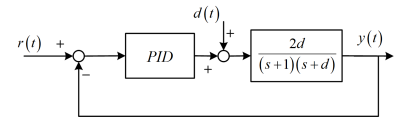


Fig. 5. PID control system

$$K_c(s) = k_c \frac{(s + c_1)(s + c_2)}{s} = P \left(1 + \frac{1}{Is} + Ds \right) \quad (17)$$

is unfortunately nonrealizable, and its realizable form is

$$K_c(s) = k_c \frac{(s + c_1)(s + c_2)}{s(\mu s + 1)} = P \left(1 + \frac{1}{Is} + \frac{Ds}{\mu s + 1} \right), \quad (18)$$

where μ is a small number. Relationships between zero-pole and classical three term form are as follows:

$$P = k_c(c_1 + c_2), \quad I = (c_1 + c_2)(c_1 c_2)^{-1}, \quad D = (c_1 + c_2)^{-1}. \quad (19)$$

These relationships are valid for $\mu = 0$, and approximately valid for $0 < \mu \ll 1$. Observe that the high frequency gain of the controller in (18) equals to k_c/μ , and for small μ it can be large. It is a common practice to consider control systems with the ideal controller, and then to use its filtered version at an implementation stage [2], [15]. As long as dynamic of a filter is negligible compared to the closed loop dynamic, the influence of μ can be neglected. Otherwise, as in our case, μ should be considered as an important design factor. A characteristic feature of the considered system is that for large k_c and small μ two modes can be distinguished. The slow mode is responsible for the disturbance response, while the fast mode, depending basically on μ , is responsible for both, the control signal that attenuates the disturbance and the reference response. Our approach consists of two stages. Firstly we choose the parameters of the ideal controller so as to obtain assumed dynamical characteristics of slow modes, and then we choose the value of μ so that the slow mode is not affected by the fast one, and that the fast mode reaches assumed dynamical characteristics.

A. Analysis of systems with an ideal PID controller

Let us consider the control system with an ideal PID controller. Relevant transfer functions are

$$G_{yd}(s) = \frac{2ds}{s(s+1)(s+d) + kd(s+c_1)(s+c_2)} \quad (20)$$

$$G_{ud}(s) = \frac{-kd(s+c_1)(s+c_2)}{s(s+1)(s+d) + kd(s+c_1)(s+c_2)}. \quad (21)$$

There is also $G_{yr}(s) = -G_{ud}(s)$. The characteristic polynomial of the closed loop system is

$$\chi(s) = s^3 + (1+d+kd)s^2 + (d+kdc_1+kdc_2)s + kdc_1c_2, \quad (22)$$

where $k = 2dk_c$. Assume that our aim is to find the controller settings so as to attain specified roots $-s_1, -s_2$ and $-s_3$ of the characteristic polynomial

$$\chi(s) = (s+s_1)(s+s_2)(s+s_3). \quad (23)$$

Comparison of (23) and (22) leads to

$$k_c = \frac{(s_1+s_2+s_3)u_0 - (d+1)}{2d} \quad (24)$$

$$c_{1,2} = \frac{(s_1s_2+s_1s_3+s_2s_3)u_0^2 - d \pm \sqrt{\Delta}}{2[(s_1+s_2+s_3)u_0 - (d+1)]} \quad (25)$$

with

$$\Delta = [(s_1s_2+s_1s_3+s_2s_3)u_0^2 - d]^2 - 4[(s_1+s_2+s_3)u_0 - (d+1)]s_1s_2s_3u_0^2. \quad (26)$$

Depending on the type of roots three cases can be distinguished:

1) A pair of complex roots and one real root : Let

$$\chi(s) = (s+\sigma+j\omega)(s+\sigma-j\omega)(s+\alpha\sigma) \quad (27)$$

$$= [s+\sigma(1+j\theta)][s+\sigma(1-j\theta)](s+\alpha\sigma) \quad (28)$$

$$= (s^2+2\sigma s+\sigma^2(1+\theta^2))(s+\alpha\sigma), \quad (29)$$

where $\theta = \omega/\sigma$. Then

$$k_c = \frac{(2+\alpha)\sigma - (d+1)}{2d} \quad (30)$$

$$c_{1,2} = \frac{\sigma^2(2\alpha+1+\theta^2) - d \pm \sqrt{\Delta}}{2[(2+\alpha)\sigma - (d+1)]} \quad (31)$$

$$\Delta = [\sigma^2(2\alpha+1+\theta^2) - d]^2 - 4[(2+\alpha)\sigma - (d+1)]\alpha\sigma^3(1+\theta^2). \quad (32)$$

For $\theta = 0$ there is $\chi(s) = (s+\sigma)^2(s+\alpha\sigma)$.

2) Different real roots:

$$\chi(s) = (s+a)(s+b)(s+c) \quad (33)$$

$$= (s+\sigma+\delta)(s+\sigma-\delta)(s+\alpha\sigma) \quad (34)$$

$$= (s^2+2\sigma s+\sigma^2(1-\phi^2))(s+\alpha\sigma) \quad (35)$$

where $\phi = \delta/\sigma \leq 1$. Then

$$k_c = \frac{(2+\alpha)\sigma - (d+1)}{2d} \quad (36)$$

$$c_{1,2} = \frac{\sigma^2(2\alpha+1-\phi^2) - d \pm \sqrt{\Delta}}{2[(2+\alpha)\sigma - (d+1)]} \quad (37)$$

$$\Delta = [\sigma^2(2\alpha+1-\phi^2) - d]^2 - 4[(2+\alpha)\sigma - (d+1)]\alpha\sigma^3(1-\phi^2). \quad (38)$$

For $\phi = 0$ there is $\chi(s) = (s+\sigma)^2(s+\alpha\sigma)$.

3) Triple root: $\chi(s) = (s+\sigma)^3$ Controller settings

$$k_c = \frac{3\sigma - (d+1)}{2d} \quad (39)$$

$$c_{1,2} = \frac{3\sigma^2 - d \pm \sqrt{\Delta}}{2[3\sigma - (d+1)]} \quad (40)$$

$$\Delta = (3\sigma^2 - d)^2 - 4[3\sigma - (d+1)]\sigma^3. \quad (41)$$

B. Disturbance responses

In this section the response $y(t)$ caused by step-wise change of $d_1(t) = 1(t)$ is analyzed depending on the character of roots.

1) Complex roots:

$$y(t) = \frac{2d[e^{-\alpha\sigma t} - e^{-\sigma t}(\cos\omega t + \frac{1-\alpha}{\theta}\sin\omega t)]}{\sigma^2[(1-\alpha)^2 + \theta^2]}. \quad (42)$$

Let us split $y(t)$ into two parts

$$y(t) = y_\alpha(t) + y_\sigma(t) \quad (43)$$

For $\alpha \geq 5$ there is

$$y(t) \approx y_\sigma(t) = \frac{-2de^{-\sigma t}(\cos\omega t + \frac{1-\alpha}{\theta}\sin\omega t)}{\sigma^2[(1-\alpha)^2 + \theta^2]}. \quad (44)$$

The maximum value y_m of $y(t)$ takes place at $t = t_m$, and there is

$$\begin{aligned} \alpha\sigma^2 y_m &= \frac{2d\alpha \left(e^{-\alpha\sigma t_m} + \sqrt{\frac{(1-\alpha)^2 + \theta^2}{1+\theta^2}} e^{-\sigma t_m} \right)}{(1-\alpha)^2 + \theta^2} \\ &\approx \frac{2de^{-\sigma t_m}}{\sqrt{1+\theta^2}}, \end{aligned} \quad (45)$$

$$\sigma t_m \approx \frac{1}{\theta} \arctan\left(\frac{\alpha\theta}{\alpha-1-\theta^2}\right) \approx \frac{\arctan(\theta)}{\theta}. \quad (46)$$

2) Real roots:

$$y(t) = \frac{d \left[2e^{-\alpha\sigma t} - e^{-\sigma t} \left(e^{\delta t} + e^{-\delta t} + \frac{1-\alpha}{\phi} (e^{\delta t} - e^{-\delta t}) \right) \right]}{\sigma^2 [(1-\alpha)^2 - \phi^2]} \quad (47)$$

Taking into account that $y(t) = y_\alpha(t) + y_\sigma(t)$ and assuming $\alpha \geq 5$ one gets

$$\alpha\sigma^2 y_m = \frac{d\alpha \left[2\Omega^{-\frac{\alpha}{2\phi}} - \Omega^{-\frac{1+\phi}{2\phi}} \left(\frac{(1-\alpha)+\phi}{\phi} \Omega - \frac{(1-\alpha)-\phi}{\phi} \right) \right]}{(1-\alpha)^2 - \phi^2}$$

$$\approx \frac{d(\Omega-1)\Omega^{-\frac{1+\phi}{2\phi}}}{\phi}, \quad (48)$$

$$\sigma t_m \approx \frac{1}{2\phi} \ln(\Omega), \quad (49)$$

where

$$\Omega = \frac{[(1-\alpha)-\phi](1+\phi)}{[(1-\alpha)+\phi](1-\phi)} \approx \frac{1+\phi}{1-\phi}. \quad (50)$$

3) Triple root:

$$y(t) = dt^2 e^{-\sigma t}, \quad \sigma^2 y_m = 4de^{-2}, \quad \sigma t_m = 2. \quad (51)$$

The influence of α, θ and ϕ on y_m for given σ is displayed in Fig.6. Observe also that t_m is proportional to the inverse of σ while y_m is proportional to the inverse of σ^2 for α, θ and ϕ unchanged.

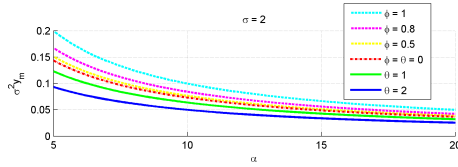


Fig. 6. Dependence of $\sigma^2 y_m$ from α . Solid line - complex roots, dotted line - real roots.

C. System with a filtered differential component

Assume now that we use the controller of (18), and we want to choose μ in such way, that the control quality will remain the same as for $\mu = 0$. For $\mu = 0$ and values of α large enough, which means k_c large enough, two roots of the characteristic equation approach c_1 and c_2 , and the third root $-\alpha\sigma$ is at the real axis in the s -plane. The degree of the system with nonzero μ equals to 4, and its relative degree equals to 2. As a result, there will be an asymptote perpendicular to the real axis crossing it at σ_a , where

$$\sigma_a = \frac{-(1+d) - \frac{1}{\mu} + c_1 + c_2}{2} \simeq \frac{-(1+d) - \frac{1}{\mu} + 2\sigma}{2} \quad (52)$$

Finally, putting $\sigma_a = -\alpha\sigma$ leads to

$$\mu = \frac{1}{2\sigma(1+\alpha) - (1+d)} \simeq \frac{1}{2\sigma\alpha} \quad (53)$$

A comparison between an ideal PID and a real PID control systems is presented in Fig.7. It is worth to notice that introduction of a filter practically does not affect the slow mode roots $s_{1,2}$

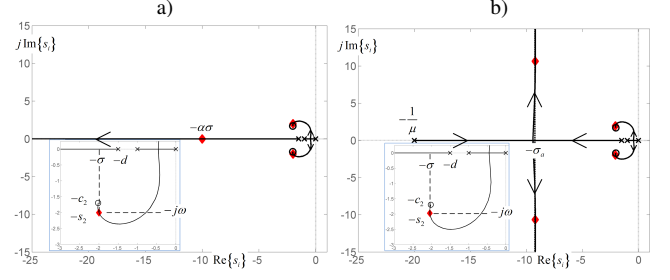


Fig. 7. PID controller for $s_{1,2} = -2(1 \pm j)$, $s_3 = -10$ ($\alpha = 5$, $\theta = 1$, $\sigma = 2$) with parameters: $k_c = 3.83$, $c_{1,2} = 2.02 \pm j1.69$. a) ideal controller $\mu = 0$, $s_{1,2} = -2 \pm 2j$, $s_3 = -10$ b) real controller $\mu = (2\sigma\alpha)^{-1} = 0.05$, $s_{1,2} = -2.04 \pm j1.97$, $s_{3,4} = -9.20 \pm j10.65$. Refer to zoom windows for details.

D. Examples

Exemplary disturbance outputs and control signals are displayed in Fig.8. It is clear that the faster the control signal approaches the value of disturbance with an opposite sign, i.e. the faster the disturbance is cancelled, the smaller the control error. Therefore, to obtain a good disturbance attenuation it is reasonable to have σ and α large. Plots of disturbance

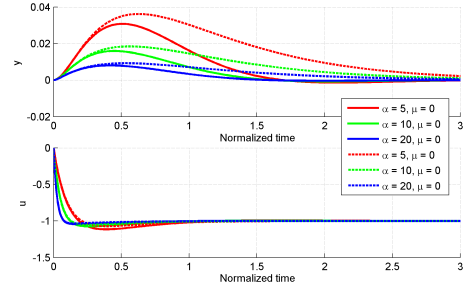


Fig. 8. Disturbance output and control for large values of α and $d = 1$, $\sigma = 2$, $\theta = 1$. Ideal PID controller with $\mu = 0$. Solid line - complex roots, dotted line - real roots.

responses for various values of μ from (53) are displayed in Fig.9. It is interesting to notice that compared to the $\mu = 0$ case of Fig.8 only the control signals have slightly changed due to replacing one real root $s_3 \simeq -\alpha\sigma$ by a complex pair $s_{3,4} \simeq -\alpha\sigma(1 \pm j)$, while the output responses remained almost the same.

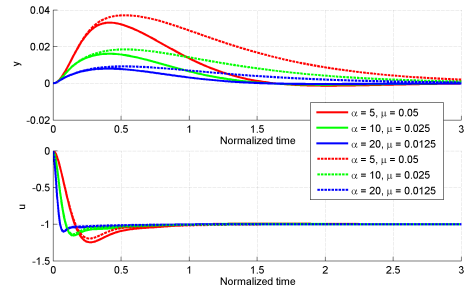


Fig. 9. Disturbance output and control for large values of α and $d = 1$, $\sigma = 2$, $\theta = 1$. Real PID controller with $\mu > 0$. Solid line - complex roots, dotted line - real roots.

For comparison with classical settings of [2], [15], plots for small values of α , and, as a result, values of k_c are displayed

in Fig.11 for different values of the working point u_o ³.

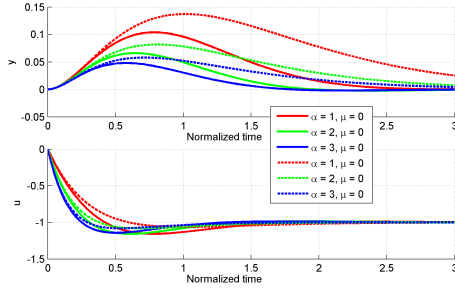


Fig. 10. Disturbance output and control for small values of α and $d = 1, \sigma = 2, \theta = 1$. Ideal PID controller with $\mu = 0$. Solid line - complex roots, dotted line - real roots.

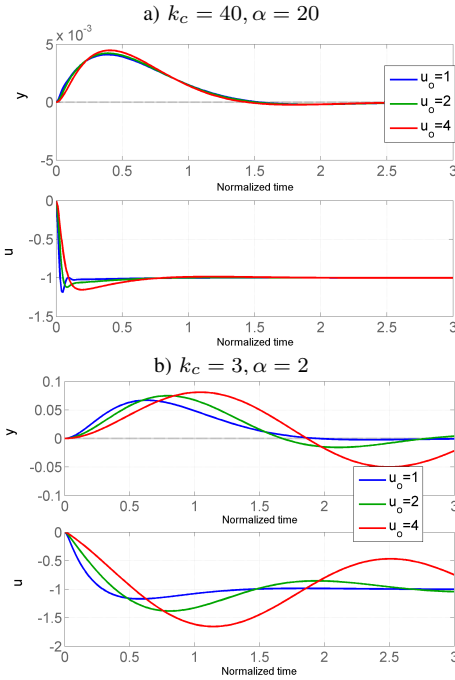


Fig. 11. Influence of the working point. PID controller with $\sigma = 2; \theta = 1, \mu = 0.0125$. a) our design, b) classical design

E. Design based on a time scale separation technique⁴

Assume $k_c = k^*/\mu$. Then

$$G_{ud}(s) = \frac{-2k^*d(s + c_1)(s + c_2)}{\mu s(\mu s + 1)(s + 1)(s + d) + 2k^*d(s + c_1)(s + c_2)}.$$

Assume a new variable $p = \mu s$ related to the fast time scale $t' = t/\mu$. Then

$$G_{ud}(p) = \frac{-2k^*d(p + \mu c_1)(p + \mu c_2)}{p(p + 1)(p + \mu)(p + \mu d) + 2k^*d(p + \mu c_1)(p + \mu c_2)}$$

³Huge progress has been done since early 80's of the previous century, when K.J. Åström and A-B Österberg [2] used 8-bit ADC's with sampling frequency of 10 Hz. Contemporary aquisition devices with 12-20 bit effective resolution and sampling frequency exceeding human hearing range allow using PID controllers with smaller μ and higher gain, which calls for new tuning rules for better control performance.

⁴The method is motivated by [13], [14], [4]

and

$$\lim_{\mu \rightarrow 0} G_{ud}(p) = \frac{-2k^*d}{p^2 + p + 2k^*d}.$$

The value of $2k^*d$ determines the roots $p_{3,4} = -1/2(1 \pm j\theta_{3,4})$ thus the shape of the control signal. In particular for $2k^*d = 1/2$ we get $\theta_{3,4} = 1$. Returning to the original time scale we get the approximate transfer function

$$G_{ud}(s) \simeq \frac{-2k^*d}{\mu^2 s^2 + \mu s + 2k^*d}$$

with $s_{3,4} = 1/(2\mu)(1 \pm j\theta_{3,4})$. For the controlled output there is

$$G_{yd}(s) = \frac{2d\mu s(\mu s + 1)}{\mu s(\mu s + 1)(s + 1)(s + d) + 2k^*d(s + c_1)(s + c_2)}$$

For μ small enough this leads to

$$G_{yd}(s) \simeq \frac{\mu s}{k^*(s + c_1)(s + c_2)} = \frac{s}{k_c(s + c_1)(s + c_2)}$$

and since for large values of k_c the poles tend to zeros there is $c_1 \simeq s_1$ and $c_2 \simeq s_2$, where $-s_1$ and $-s_2$ are the required poles.

From these considerations the following simple design procedure results: Choose the values of α, μ and σ such that $\mu\sigma = 1/(2\alpha)$, where $\alpha \geq 5$ determines the time scale separation, set $c_{1,2} = \sigma(1 \pm j\theta)$ and $k_c = k^*/\mu = \frac{1}{4d\mu}$. As a result, taking (19) into account there is

$$P \simeq \frac{\alpha\sigma^2}{d} = \frac{1}{4\alpha d\mu^2}, \quad (54)$$

$$I \simeq \frac{2}{\sigma(1 + \theta^2)} = \frac{4\alpha\mu}{1 + \theta^2}, \quad (55)$$

$$D \simeq \frac{1}{2\sigma} = \alpha\mu. \quad (56)$$

IV. REFERENCE AND FEEDFORWARD SIGNALS GENERATORS

If the value of controller gain is high, and the derivative part detects fast changes of the control error, the control system is prone to saturation. Therefore a controller with the anti-windup augmentation, see e.g. Fig.12, is necessary. However, as indicated in Fig.15.a, for a wider range of working points and reference changes, the behavior of the control system is not fully predictable. In particular, one can observe overshoots and large changes of water level in the first tank, which can result in overflow. As shown in Fig.16, anti-windup can also be activated by disturbance $d_2(t)$ acting in the lower tank. A reasonable solution to prevent saturation is to deliver the reference signal which the control system is able to follow smoothly without exceeding the control signal limitations. This can be done by passing the step-wise reference through a second order reference filter

$$K^{ref}(s) = \frac{a_0^r}{s^2 + a_1^r s + a_0^r} \quad (57)$$

with appropriately chosen parameters. Finally, a feedforward signal calculated from the setpoint can facilitate the reference following.

V. CONCLUSION

In this paper⁵, PID control of a cascade tanks system was investigated. We proposed a very simple method of controller parameter settings that bases on asymptotic properties of the control system for larger values of controller gain and faster filter of the controller differential term than those used in classical approach. This results in two time-scale behavior of the controlled system with very fast control signal and slower error signal. We showed that the control error caused by the disturbance in the upper tank can be reduced to extremely small values, and that it is invariant to working point changes. As a result, we obtained better performance of control than that based on known literature. The method of controller synthesis is very simple and intuitive, and as such very well suited for introductory control course based either on laboratory experiment or on virtual interactive control course [11].

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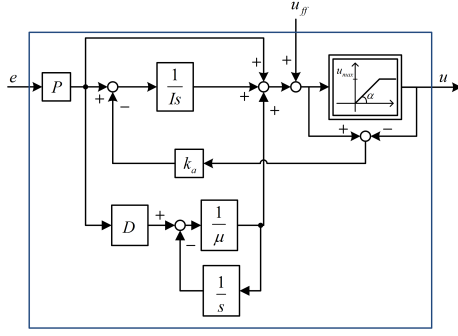


Fig. 12. PID with anti-windup and feedforward signal input

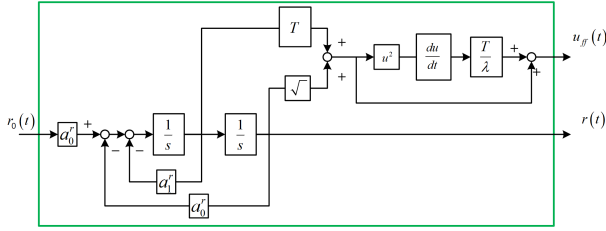


Fig. 13. Second order set-point and feedforward generator

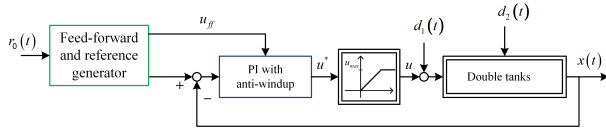


Fig. 14. Feedback and feedforward control system

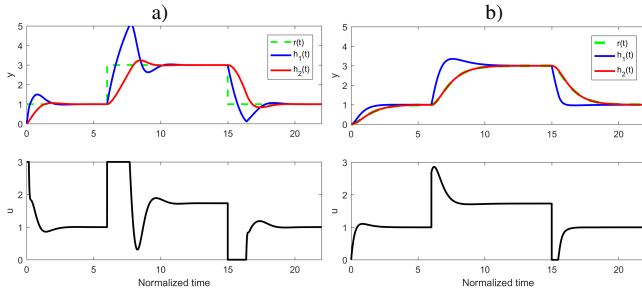


Fig. 15. a) PID with antiwindup and stepwise reference changes, b) 2nd order reference model and PID with antiwindup; $K^{ref}(s) = 1.69/(s + 1.3)^2$

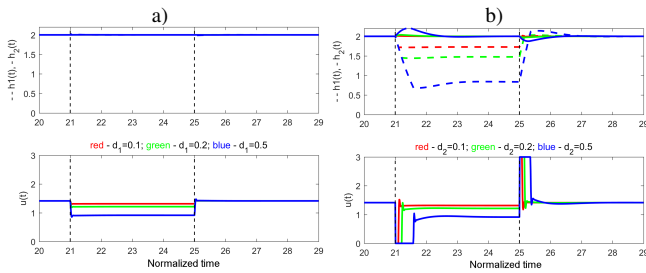


Fig. 16. Effect of disturbance acting in the upper tank (a) and the lower tank (b)

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